MERCK MANUAL
Professional Version

Hypothyroidism

(Myxedema)

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Hypothyroidism is thyroid hormone deficiency. Symptoms include cold intolerance, fatigue, and weight gain. Signs may include a typical facial appearance, hoarse slow speech, and dry skin. Diagnosis is with thyroid function tests. Management includes administration of thyroxine.

Hypothyroidism occurs at any age but is particularly common among older adults, where it may present subtly and be difficult to recognize. Hypothyroidism may be primary (caused by disease in the thyroid) or secondary (caused by disease in the hypothalamus or pituitary). (See also Overview of Thyroid Function.)

Primary hypothyroidism

Primary hypothyroidism is due to decreased secretion of thyroxine (T4) and triiodothyronine (T3) from the thyroid. Serum T4 and T3 levels are low, and thyroid-stimulating hormone (TSH) level is increased. In the United States, the most common cause is autoimmune inflammation. It usually results from Hashimoto thyroiditis and is often associated with a firm goiter or, later in the disease process, with a shrunken fibrotic thyroid with little or no function.

The second most common cause is treatment for hyperthyroidism (post-therapeutic hypothyroidism), especially after **radioactive iodine therapy** or surgery for hyperthyroidism, goiter, or thyroid cancer. Hypothyroidism during overtreatment with propylthiouracil, methimazole, or iodide abates after therapy is stopped.

Iodine deficiency may cause endemic goiter and goitrous hypothyroidism. Most patients with goiters not caused by Hashimoto thyroiditis are euthyroid or have hyperthyroidism. Iodine deficiency decreases thyroid hormonogenesis. In response, TSH is released, which causes the thyroid to enlarge and trap iodine avidly; thus, goiter results. If iodine deficiency is severe, the patient becomes hypothyroid, a rare occurrence in the most parts of the world s since the advent of iodized salt.

Iodine deficiency can cause congenital hypothyroidism. In severely iodine-deficient regions worldwide, congenital hypothyroidism (previously termed endemic cretinism) is a major cause of intellectual disability.

Rare inherited **enzymatic defects** can alter the synthesis of thyroid hormone and cause goitrous hypothyroidism.

Hypothyroidism may occur in patients taking lithium, perhaps because lithium inhibits hormone release by the thyroid. Hypothyroidism may also occur in patients taking amiodarone or other iodine-containing medications, in patients taking interferon-alfa, and in patients being treated for cancer with checkpoint inhibitors or some tyrosine kinase inhibitors.

Hypothyroidism can result from radiation therapy for cancer of the larynx or Hodgkin lymphoma. The incidence of permanent hypothyroidism after radiation therapy is high, and thyroid function (through measurement of serum TSH) should be evaluated at 6- to 12-month intervals.

Secondary hypothyroidism

Secondary hypothyroidism occurs when the hypothalamus produces insufficient thyrotropin-releasing hormone (TRH) or the pituitary produces insufficient TSH.

Sometimes, deficient TSH secretion due to deficient TRH secretion is termed tertiary hypothyroidism.

Subclinical hypothyroidism

Subclinical hypothyroidism is elevated serum TSH in patients with absent or minimal symptoms of hypothyroidism and normal serum levels of free thyroxine (T4).

Subclinical thyroid dysfunction is relatively common; it occurs in approximately 15% of older women and 10% of older men (1, 2), particularly in those with underlying Hashimoto thyroiditis.

In patients with serum TSH > 10 mIU/L (> 10 microIU/mL), there is a high likelihood of progression to overt hypothyroidism with low serum levels of free T4 within the next 10 years. These patients are also more likely to have hypercholesterolemia and atherosclerosis. They should be treated with levothyroxine, even if they are asymptomatic.

For patients with TSH levels between 4.5 and 10 mUI/L, (4.5 and 10 microIU/L) a trial of levothyroxine is reasonable if symptoms of early hypothyroidism (eg, fatigue, depression) are present.

Levothyroxine therapy is also indicated in pregnant women and in women who plan to become pregnant to avoid deleterious effects of hypothyroidism on the pregnancy and fetal development. Patients should have annual measurement of serum TSH and free T4 to assess progress of the condition if untreated or to adjust the levothyroxine dosage.

General references

1. Cappola AR, Fried LP, Arnold AM, et al. Thyroid status, cardiovascular risk, and mortality in older adults. *JAMA.* 2006;295(9):1033-1041. doi:10.1001/jama.295.9.1033
2. Jasim S, Papaleontiou M. Considerations in the Diagnosis and Management of Thyroid Dysfunction in Older Adults. *Thyroid.* 2025;35(6):624-632. doi:10.1089/thy.2025.0128

Symptoms and Signs of Hypothyroidism

Symptoms and signs of primary hypothyroidism are often subtle and insidious. The most common presenting symptoms are fluid retention and puffiness, especially periorbitally, tiredness, cold intolerance, and mental fogginess.

Thyroid and Parathyroid Glands

Various organ systems may be affected with many possible signs and symptoms, including:

- Metabolic: Cold intolerance, modest weight gain (due to fluid retention and decreased metabolism), hypothermia
- Neurologic: Forgetfulness, paresthesias of the hands and feet (often due to carpal tunnel syndrome caused by deposition of proteinaceous ground substance in the ligaments around the wrist and ankle); slowing of the relaxation phase of deep tendon reflexes
- Psychiatric: Personality changes, depression, dull facial expression, dementia or frank psychosis (myxedema madness)
- Dermatologic: Facial puffiness; myxedematous swelling and thickening of the skin; sparse, coarse and dry hair; coarse, dry, scaly, and thick skin; carotenemia, particularly notable on the palms and soles (caused by deposition of carotene in the lipid-rich epidermal layers); macroglossia due to deposition of proteinaceous ground substance in the tongue
- Ocular: Periorbital swelling due to infiltration with the mucopolysaccharides hyaluronic acid and chondroitin sulfate, droopy eyelids because of decreased adrenergic drive
- Gastrointestinal: Constipation
- Gynecologic: Heavy menstrual bleeding or secondary amenorrhea
- Cardiovascular: Slow heart rate (a decrease in both thyroid hormone and adrenergic stimulation causes bradycardia), enlarged heart on examination and imaging (partly because of dilation but chiefly because of pericardial effusion; pericardial effusions develop slowly and only rarely cause hemodynamic distress)
- Other: Pleural or abdominal effusions (pleural effusions develop slowly and only rarely cause respiratory or hemodynamic distress), hoarse voice, and slow speech

Myxedema in Hypothyroidism

IMAGE

BY PERMISSION OF THE PUBLISHER. FROM BURMAN K, BECKER K, CYTRYN A, ET AL. IN *ATLAS OF CLINICAL ENDOCRINOLOGY: THYROID DISEASES*. EDITED BY SG KORENMAN (SERIES EDITOR) AND MI SURKS. PHILADELPHIA, CURRENT MEDICINE, 1999.

Symptoms can differ significantly in older patients.

Although secondary hypothyroidism is uncommon, its causes often affect other endocrine organs controlled by the hypothalamic-pituitary axis. In a woman with hypothyroidism, indications of secondary hypothyroidism are a history of amenorrhea rather than heavy menstrual bleeding and some suggestive differences on physical examination.

Secondary hypothyroidism is characterized by skin and hair that are dry but not very coarse, skin depigmentation, only minimal macroglossia, atrophic breasts, and low blood pressure. Also, the heart is small, and serous pericardial effusions do not occur. Hypoglycemia is common because of concomitant adrenal insufficiency or growth hormone deficiency.

Myxedema coma

Myxedema coma is a life-threatening complication of hypothyroidism, usually occurring in patients with a long history of hypothyroidism. Its characteristics include coma with extreme hypothermia (temperature 24° to 32.2° C), areflexia, seizures, and respiratory depression with carbon dioxide retention. Severe hypothermia may be missed unless low-reading thermometers are used. Rapid diagnosis based on clinical judgment, history, and physical examination is imperative, because death is likely without rapid treatment. Precipitating factors include illness, infection, trauma, medications or substances that suppress the central nervous system, and exposure to cold.

Diagnosis of Hypothyroidism

- Thyroid-stimulating hormone (TSH) level

- Free thyroxine (T4) level

Serum thyroid-stimulating hormone measurement is the most sensitive test for diagnosing hypothyroidism. In primary hypothyroidism, there is decreased feedback inhibition of the intact pituitary; serum TSH is always elevated, whereas serum free T4 is low. In secondary hypothyroidism, free T4 and serum TSH are low (sometimes TSH is normal but with decreased bioactivity).

Many patients with primary hypothyroidism have normal circulating levels of triiodothyronine (T3), probably caused by sustained TSH stimulation of the failing thyroid, resulting in preferential synthesis and secretion of biologically active T3. Therefore, serum T3 is not sensitive for hypothyroidism.

Anemia is often present, usually normocytic-normochromic and of unknown etiology, but it may be hypochromic because of iron deficiency due to heavy menstrual bleeding and sometimes macrocytic because of associated pernicious anemia or decreased absorption of folate. Anemia is rarely severe (hemoglobin is usually > 9 g/dL [> 90 g/L]). As the hypometabolic state is corrected, anemia subsides, sometimes requiring 6 to 9 months.

Serum cholesterol is usually high in primary hypothyroidism but not as high in secondary hypothyroidism.

In addition to primary and secondary hypothyroidism, other conditions may cause decreased levels of total T4, such as euthyroid sick syndrome and serum thyroxine-binding globulin (TBG) deficiency.

Screening

Recommendations regarding screening for hypothyroidism and thyroid dysfunction in general vary among professional organizations (1, 2, 3). Screening for hypothyroidism is warranted in select populations (eg, neonates, older adults with risk factors) in which it is relatively more prevalent, especially because it can cause significant morbidity and its manifestations can be subtle. Screening is done by measuring TSH levels.

Diagnosis references

1. Croswell J, Ballard T. Screening for Thyroid Dysfunction. *Am Fam Physician.* 2015;92(8):717-718.
2. Garber JR, Cobin RH, Gharib H, et al. Clinical practice guidelines for hypothyroidism in adults: cosponsored by the American Association of Clinical Endocrinologists and the American Thyroid Association. *Endocr Pract.* 2012;18(6):988-1028. doi:10.4158/EP12280.GL
3. LeFevre ML; U.S. Preventive Services Task Force. Screening for thyroid dysfunction: U.S. Preventive Services Task Force recommendation statement. *Ann Intern Med.* 2015;162(9):641-650. doi:10.7326/M15-0483

Treatment of Hypothyroidism

- Levothyroxine, adjusted until TSH levels are in midnormal range (1)

Various thyroid hormone preparations are available for replacement therapy, including synthetic preparations of T4 (levothyroxine), T3 (liothyronine), combinations of the 2 synthetic hormones, and desiccated porcine thyroid extract. Levothyroxine is preferred; the usual maintenance dose is 75 to 150 mcg orally once a day, depending on age, body mass index, and absorption (for pediatric treatment, see Hypothyroidism in Infants and Children). In young or middle-aged patients who are otherwise healthy, the starting dose of levothyroxine can be 100 mcg or 1.7 mcg/kg orally once a day.

In patients with heart disease, therapy is begun with low doses of levothyroxine, usually 25 mcg once a day. The dose is adjusted every 6 weeks until maintenance dose is achieved. The maintenance dose may need to be increased in patients who are pregnant. Dose may also need to be increased if medications that decrease T4 absorption (eg, calcium, iron, magnesium, cholestyramine) or increase its metabolic clearance (eg, rifampin, carbamazepine, phenytoin, phenobarbital, topiramate) are administered concomitantly. The dose used should be the lowest that restores serum TSH levels to the midnormal range (though this criterion cannot be used in patients with secondary hypothyroidism). In secondary hypothyroidism the dose of levothyroxine should achieve a free T4 level in the midnormal range.

Liothyronine (L-triiodothyronine) should not be used alone for long-term replacement because of its short half-life and the large peaks in serum T3 levels it produces. The administration of standard replacement amounts (25 to 37.5 mcg twice a day) results in rapidly increasing serum T3 to between 300 and 1000 ng/dL (4.62 to 15.4 nmol/L) within 4 hours due to its almost complete absorption; these levels return to normal by 24 hours. Additionally, patients receiving liothyronine are chemically hyperthyroid for at least several hours a day, potentially increasing cardiac risks. T3 does not cross the placenta and should not be administered to patients who are pregnant.

Similar patterns of serum T3 changes occur when mixtures of T3 and T4 are taken orally, although peak T3 is lower because less T3 is given. Replacement regimens with synthetic T4 preparations reflect a different pattern in serum T3 response. Increases in serum T3 occur gradually, and normal levels are maintained when adequate doses of T4 are given. Desiccated animal thyroid preparations contain variable amounts of T3 and T4 and should not be prescribed unless the patient is already taking the preparation and has normal serum TSH.

In patients with secondary hypothyroidism, levothyroxine should not be given until there is evidence of adequate cortisol secretion (or cortisol therapy is given), because levothyroxine could precipitate adrenal crisis.

Myxedema coma

Myxedema coma is treated as follows:

- T4 given IV
- Glucocorticoids
- Supportive care as needed
- Conversion to oral T4 when patient is stable

Patients require a large initial dose of T4 (300 to 500 mcg IV) or T3 (25 to 50 mcg IV). The intravenous maintenance dose of T4 is 75 to 100 mcg once a day and of T3, 10 to 20 mcg twice a day until T4 can be given orally. Glucocorticoids are also given because the possibility of central hypothyroidism and central adrenal insufficiency usually cannot be initially excluded. The patient should not be rewarmed rapidly, which may precipitate hypotension or arrhythmias.

Hypoxemia is common, so the partial pressure of oxygen in arterial blood (PaO2) should be monitored. If ventilation is compromised, immediate mechanical ventilatory assistance is required. The precipitating factor should be rapidly and appropriately treated and fluid replacement given carefully, because patients who are hypothyroid do not excrete water appropriately. Finally, all medications should be given cautiously because they are metabolized more slowly than in healthy people.

Treatment reference

1. Jonklaas J, Bianco AC, Bauer AJ, et al. Guidelines for the Treatment of Hypothyroidism: Prepared by the American Thyroid Association Task Force on Thyroid Hormone Replacement. *Thyroid.* 2014;24(12): 1670-1751. doi: 10.1089/thy.2014.0028

Geriatrics Essentials: Hypothyroidism

Hypothyroidism is particularly common among older adults. It occurs in close to 10% of women and 6% of men > 65 years. Although typically easy to diagnose in younger adults, hypothyroidism may be subtle and manifest atypically in older adults.

Older patients have significantly fewer symptoms than do younger patients, and symptoms and signs are often subtle and vague. Many older patients with hypothyroidism present with nonspecific geriatric syndromes—confusion, anorexia, weight loss, falling, incontinence, and decreased mobility. Musculoskeletal symptoms (especially arthralgias) occur often, but arthritis is rare. Muscular aches and weakness, often mimicking polymyalgia rheumatica or polymyositis, and an elevated creatine kinase (CK) level may occur. In older patients, hypothyroidism may mimic dementia or parkinsonism.

In older patients, levothyroxine therapy is begun with low doses, usually 25 mcg once a day. Maintenance doses may also need to be lower in older patients.

Key Points

- **Hypothyroidism is deficiency of thyroid hormone.**
- **The most common etiology of hypothyroidism is primary thyroid disease, often due to Hashimoto thyroiditis, but sometimes caused by some treatments (thyroid surgery, radioactive iodine therapy, radiation therapy to the neck, or some medications).**
- **Secondary hypothyroidism is caused by hypothalamic or pituitary disease.**
- **Signs and symptoms of primary hypothyroidism are often mild and nonspecific and can include fatigue, cold intolerance, constipation, fluid retention, periorbital edema, and/or impaired mental clarity.**

- Subclinical hypothyroidism (minimal or no symptoms and an elevated serum level of thyroid-stimulating hormone [TSH]) and normal free thyroxine [T4]) is common.
- Diagnose primary or secondary hypothyroidism by testing serum TSH and T4. In primary hypothyroidism, TSH is elevated and T4 is low; in secondary hypothyroidism, both TSH and T4 are low.
- Myxedema coma (coma, hypothermia, and other organ dysfunction) is a rare life-threatening condition that can occur during acute illness in patients with hypothyroidism; it requires rapid diagnosis and treatment.
- Treat with levothyroxine and adjust dose until TSH levels are normal.
- Check cortisol levels in patients with secondary hypothyroidism before starting thyroid replacement therapy, because of the serious risk of adrenal crisis.
- Screen selected patients (neonates, older adults with risk factors) with a serum TSH.

Drug Information for the Topic